Unit-3

IOT Data link layer and Network layer protocols

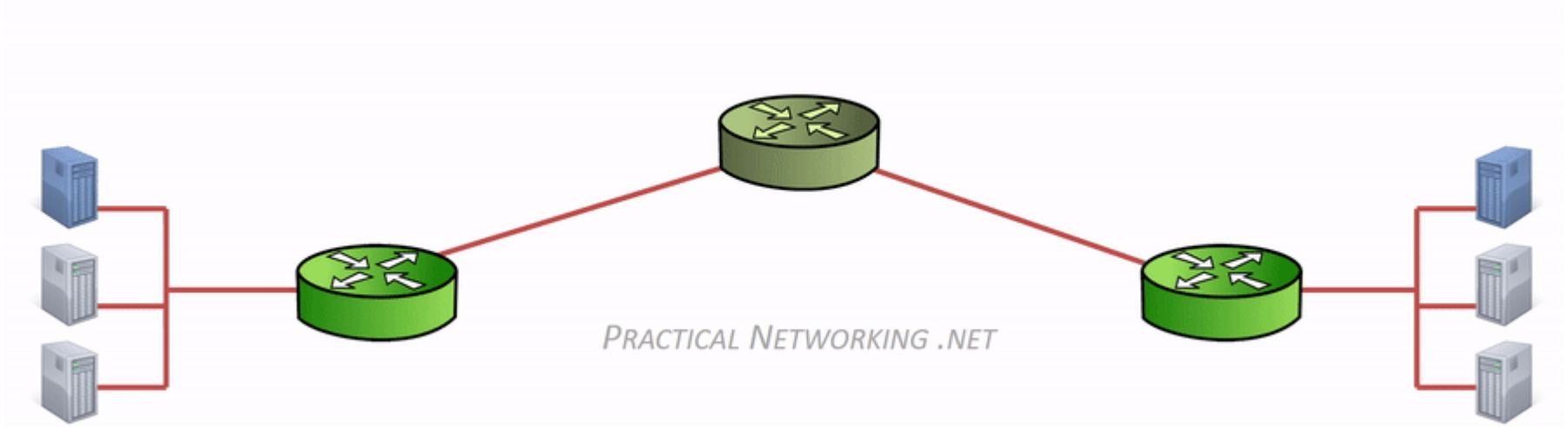
- PHY/MAC Layer
  - 3GPP MTC
  - IEEE 802.11
  - IEEE 802.15
  - Wireless HART
  - Z Wave
  - Bluetooth Low Energy
  - Zigbee Smart Energy
  - DASH7

- Network Layer
  - IPv4
  - IPv6
  - 6LoWPAN
  - 6TiSCH
  - ND
  - DHCP
  - ICMP
  - RPL
  - CORPL
  - CARP

Unit-3

IOT Data link layer and Network layer protocols

Network Layer



ROADMAP

Lecture Outline

1 

The Network Layer in the IoT Stack

2 

IPv4 — Addressing & Header Format

3 

IPv6 — Addressing & Header Format

4 

IPv4 vs IPv6 for IoT Networks

5 

6LoWPAN — Adapting IPv6 for LLNs

6 

Applications & Key Takeaways

WHERE IT FITS

The Network Layer in the IoT Protocol Stack

Application Layer	HTTP, CoAP, MQTT — data the user cares about
Transport Layer	UDP, TCP — end-to-end delivery
Network Layer	IPv4 / IPv6 / 6LoWPAN — addressing & routing
Link Layer	IEEE 802.15.4, Wi-Fi, BLE — framing on a link
Physical Layer	Radio, cable — raw bit transmission



Highlighted layer is today's focus — where packets get an address and a path.

FUNDAMENTALS

Role & Challenges of the Network Layer



Key Functions

- Logical addressing of every device
- Routing packets across multiple hops
- Fragmentation & reassembly of packets
- Interoperability across heterogeneous links



IoT Constraints

- Tiny frame sizes (e.g. 127 bytes in 802.15.4)
- Limited memory, CPU and battery power
- Lossy, low-bandwidth wireless links
- Billions of devices need unique addresses

NETWORK LAYER · LEGACY PROTOCOL

IPv4 — Overview & Addressing

- 32-bit address, written in dotted-decimal notation
- Provides about 4.3 billion unique addresses
- Addresses grouped into classes (A, B, C, D, E)
- Still the dominant protocol on the global Internet
- Relies heavily on NAT to cope with address scarcity

Dotted-Decimal Example

192 . 168 . 001 . 010

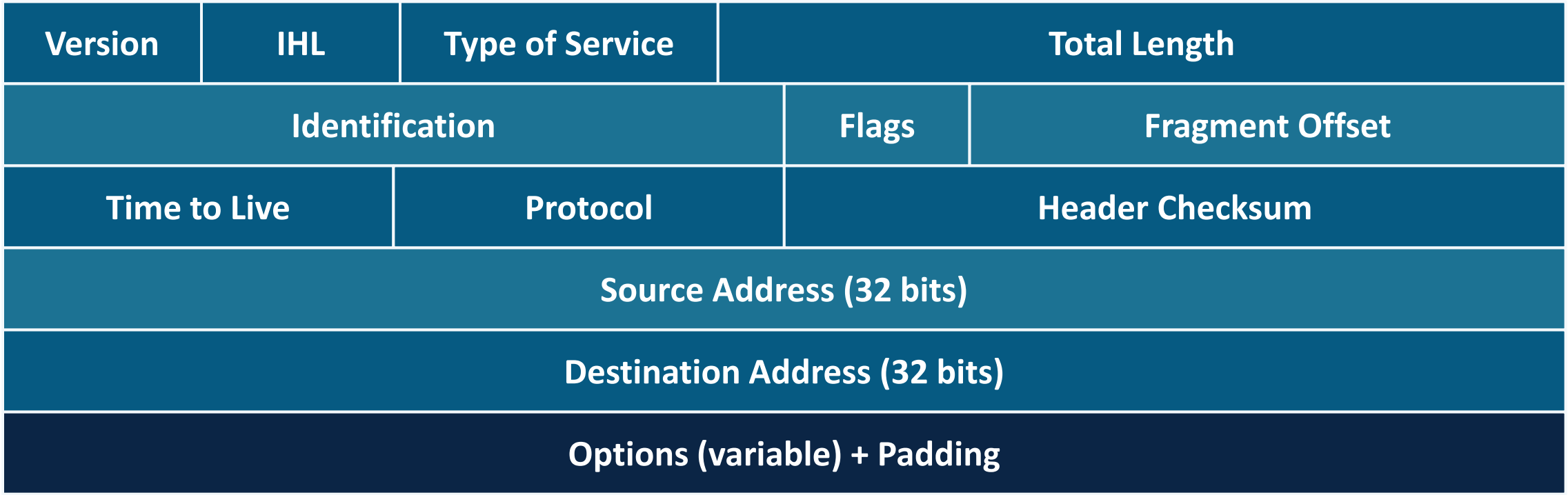
Each of the 4 octets = 8 bits → 32 bits total

Range per octet: 0 – 255

PACKET STRUCTURE

IPv4 Header Format

20-byte fixed header (RFC 791), extendable with options



THE PROBLEM

Why IPv4 Falls Short for IoT



Address Exhaustion

~4.3B addresses cannot uniquely tag tens of billions of IoT devices.



NAT Complexity

Address sharing via NAT breaks end-to-end connectivity & adds overhead.



Large Header Overhead

20+ byte header is costly on constrained, low-bandwidth radio links.

IPv6 — Overview & Addressing

- 128-bit address, written in 8 groups of hex digits
- Provides about 3.4 × 10^38 unique addresses
- Supports stateless address autoconfiguration (SLAAC)
- Simplified, fixed-size 40-byte base header
- Security (IPsec) designed in, not bolted on

Hexadecimal Example

2001	0DB8	85A3	0000
0000	8A2E	0370	7334

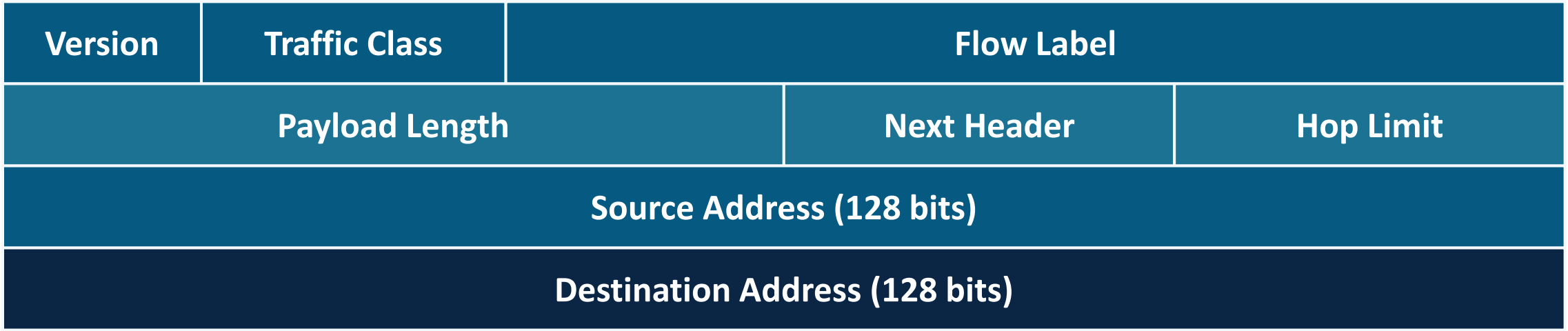
8 groups × 16 bits = 128 bits total

Consecutive zero groups may be shortened with "::"

PACKET STRUCTURE

IPv6 Header Format

Fixed 40-byte base header — simpler & faster to process than IPv4



No header checksum, no built-in fragmentation fields — extension headers handle optional features instead.

IPv6 Header Format

Field	Length	Description
Version	4 bits	IPv6 version (always `6`).
Traffic Class	8 bits	Used for differentiating between various types of traffic (e.g., for QoS).
Flow Label	20 bits	Used to identify packets belonging to the same flow.
Payload Length	16 bits	The length of the payload (data) following the header, in bytes.
Next Header	8 bits	Specifies the type of header immediately following the IPv6 header (e.g., TCP, UDP, ICMPv6).
Hop Limit	8 bits	Similar to TTL in IPv4, it determines the maximum number of hops a packet can make.
Source Address	128 bits	The IPv6 address of the sender.
Destination Address	128 bits	The IPv6 address of the recipient.

CAPABILITIES

IPv6 Address Types & Key Features



Unicast

One packet → one interface



Multicast

One packet → a group of interfaces



Anycast

One packet → nearest interface in a group

Why these matter for IoT

- Every sensor/actuator gets a globally unique address — no NAT needed
- SLAAC lets devices self-configure on joining a network
- Multicast efficiently pushes updates to many devices at once

HEAD TO HEAD

IPv4 vs IPv6 for IoT Networks

Attribute	IPv4	IPv6
Address length	32 bits	128 bits
Address space	~4.3 billion	~3.4 × 10^38
Header size	20+ bytes (variable)	40 bytes (fixed)
Configuration	Manual / DHCP	Autoconfiguration (SLAAC)
Security	Optional (add-on)	Built-in (IPsec)
Fit for massive IoT scale	Limited — needs NAT	Native, address-rich

TAKEAWAY

Why IPv6 Matters for the Internet of Things



Room for every device

Enough addresses for billions of sensors, actuators & gateways — no address reuse tricks needed.



Simpler operation

Stateless autoconfiguration lets constrained devices join a network with minimal setup traffic.



A foundation to build on

Its address model is exactly what adaptation layers like 6LoWPAN are designed to carry.

ADAPTATION LAYER

Introduction to 6LoWPAN

- IPv6 over Low-power Wireless Personal Area Networks
- Standardised in RFC 4944 and refined in RFC 6282
- Lets tiny, battery-powered devices run full IPv6
- Bridges a key mismatch: IPv6 needs a 1280-byte MTU, but IEEE 802.15.4 frames are only up to 127 bytes
- Sits between the IPv6 and the link layer in the stack



Low-power & Lossy Network (LLN)

Battery-powered sensor nodes, short radio range, high packet loss — exactly the conditions 6LoWPAN is built for.

WHERE IT SITS

The 6LoWPAN Protocol Stack

Application Layer	CoAP, MQTT-SN
Transport Layer	UDP
IPv6	Network layer addressing
6LoWPAN Adaptation Layer	Compression, fragmentation, mesh routing
IEEE 802.15.4 MAC	Channel access, framing
IEEE 802.15.4 PHY	Radio transmission (2.4 GHz)

CORE MECHANISMS

6LoWPAN Adaptation Layer — Key Functions

Header Compression

Shrinks IPv6/UDP headers (HC1, then IPHC) to fit small 802.15.4 frames.

Fragmentation & Reassembly

Splits IPv6 packets larger than 127 bytes across multiple link-layer frames.

Mesh Addressing & Routing

Forwards packets hop-by-hop (mesh-under) or via IP routing (route-over).

Neighbor Discovery Optimization

Reduces multicast-heavy IPv6 discovery traffic for low-power nodes.

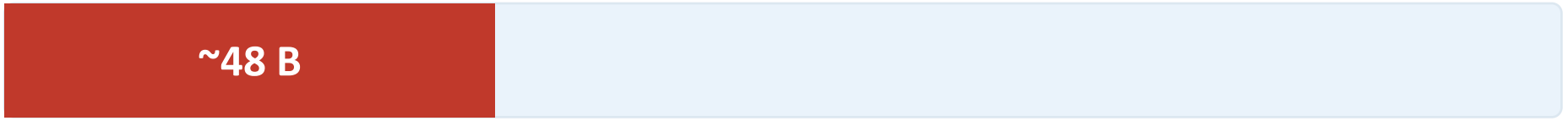
WHY COMPRESSION MATTERS

6LoWPAN Header Compression

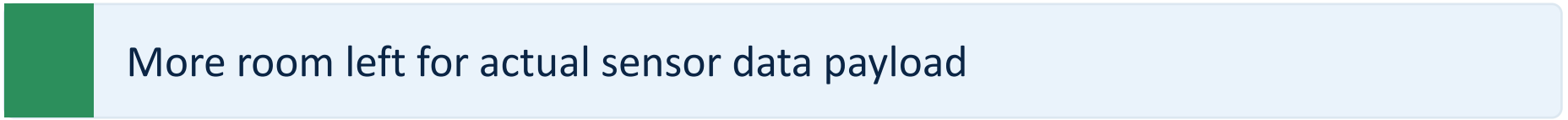
An IEEE 802.15.4 frame holds at most 127 bytes — headers must be squeezed to leave room for data

IEEE 802.15.4 frame — 127 bytes total

Uncompressed IPv6 + UDP header ≈ 48 bytes



Compressed with 6LoWPAN IPHC — as low as ~2–6 bytes



Common fields (version, traffic class, flow label...) are inferred from context instead of being sent — only what changes is transmitted.

IN THE REAL WORLD

Applications of IPv6 / 6LoWPAN in IoT



Smart Homes

Lighting, thermostats & plugs on one mesh



Industrial IoT

Thousands of sensors monitored on factory floors



Smart Agriculture

Soil & climate sensors across large fields



Healthcare Wearables

Low-power body-area monitoring devices



Smart Cities

Streetlights, parking & utility metering at scale

WRAP-UP

Summary & Key Takeaways



IPv4 gave the Internet its start, but its 32-bit space cannot fit the scale of IoT.



IPv6's 128-bit address space and simpler header make it the natural fit for billions of devices.



6LoWPAN adapts IPv6 for constrained, low-power wireless links through compression, fragmentation and mesh routing.



Together, IPv6 + 6LoWPAN form the network-layer foundation of most modern IoT deployments.

ROADMAP

6TiSCH, NDP

01

IoT Protocol Stack Recap

Where the network layer sits between MAC and applications

02

Challenges of LLNs

Why constrained, lossy links demand new protocol thinking

03

6TiSCH Fundamentals

IPv6 meets deterministic TSCH scheduling for industrial IoT

04

6top / 6P & Scheduling

Negotiating and securing communication cells

05

IPv6 Neighbor Discovery

Classic ND, and why it breaks down in mesh networks

06

6LoWPAN ND

Registration-based discovery built for constrained meshes

FOUNDATIONS

Where the Network Layer Fits

Application	CoAP · MQTT-SN
Transport	UDP
Network	IPv6 · RPL · 6LoWPAN ND · 6TiSCH
MAC / Link	IEEE 802.15.4e (TSCH)
Physical	IEEE 802.15.4 PHY



The network layer glues addressing, routing and scheduling together. 6TiSCH brings deterministic, low-power scheduling to IPv6 meshes; Neighbor Discovery lets constrained nodes find and register with routers efficiently.

Why Low-Power, Lossy Networks Need New Thinking

Tight Energy Budgets

Battery-powered nodes must sleep aggressively; radios dominate power use.

Lossy, Unstable Links

Interference and multipath fading cause frequent packet loss.

Constrained Resources

Kilobytes of RAM/ROM rule out heavyweight IPv6 mechanisms.

Large, Dynamic Meshes

Hundreds of multihop nodes join, move, and leave over time.

These constraints are exactly what 6TiSCH (scheduling) and 6LoWPAN ND (discovery) are designed to solve.

6TISCH

What Is 6TiSCH — and Why It Exists

6TiSCH = IPv6 over the TSCH mode of IEEE 802.15.4e. An IETF working group that connects deterministic, channel-hopping radio scheduling with the standard IPv6 network stack used across the Internet of Things.

- Bring industrial-grade reliability (>99.9%) to wireless mesh links
- Cut energy use with synchronized sleep/wake radio duty cycling
- Reuse standard IPv6, RPL and 6LoWPAN — no new network stack
- Allow schedules to adapt to traffic, topology and interference

Built For

- Industrial automation
- Process & asset monitoring
- Smart grid & critical infrastructure

Standardized by the IETF 6TiSCH WG, layered on IEEE 802.15.4e TSCH.

6TISCH

The 6TiSCH Protocol Architecture

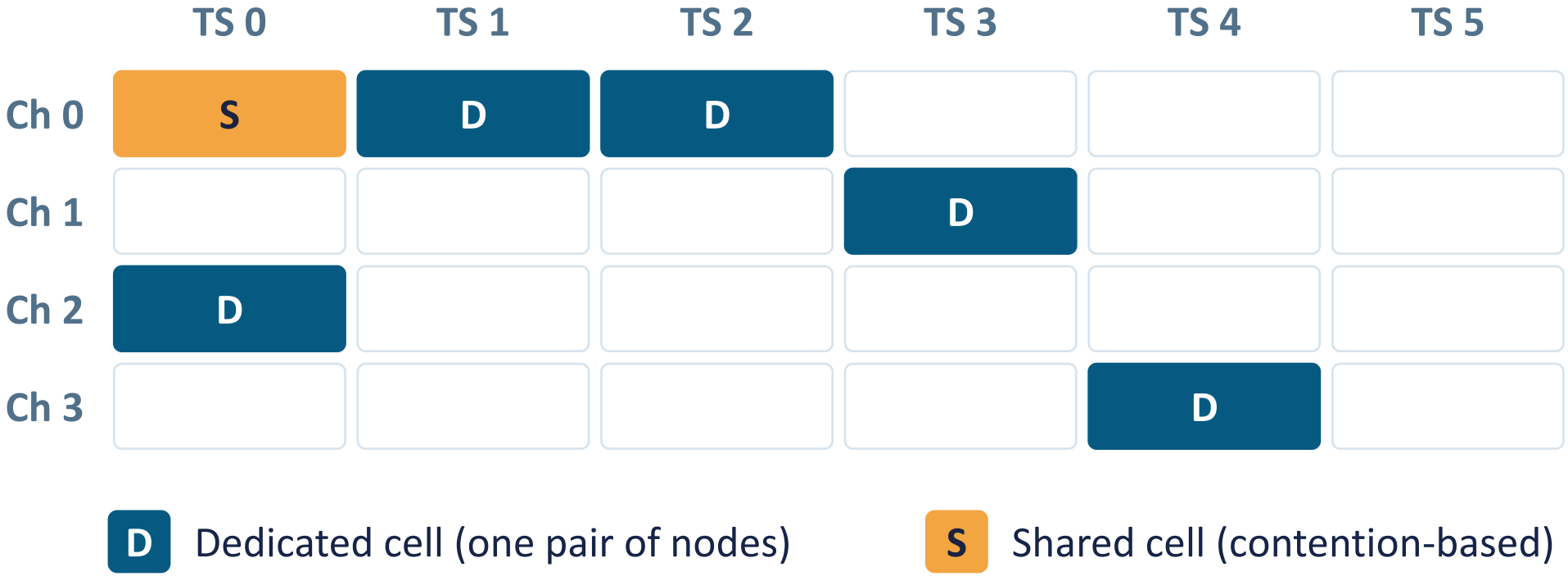
Application / CoAP	End-user IoT services & management
IPv6 · RPL · 6LoWPAN ND	Routing, addressing & neighbor discovery
6top Sublayer (6P Protocol)	Negotiates cells between neighbors
IEEE 802.15.4e — TSCH MAC	Time-slotted, channel-hopping schedule
IEEE 802.15.4 PHY	Physical radio layer

6TiSCH slots into the network + MAC boundary: IPv6/RPL above, TSCH scheduling below.

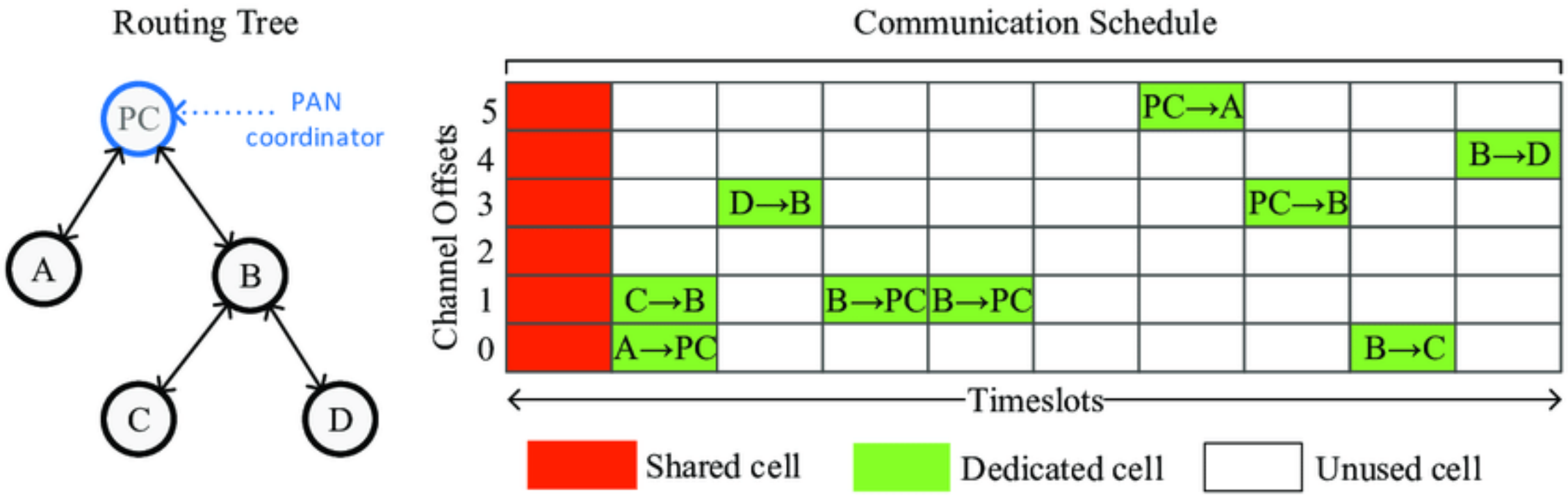
6TISCH

TSCH — Time-Slotted Channel Hopping

A slotframe repeats over time: each cell is a (timeslot, channel offset) pair. Nodes wake only for their assigned cells, then sleep — saving energy while hopping channels to dodge interference.



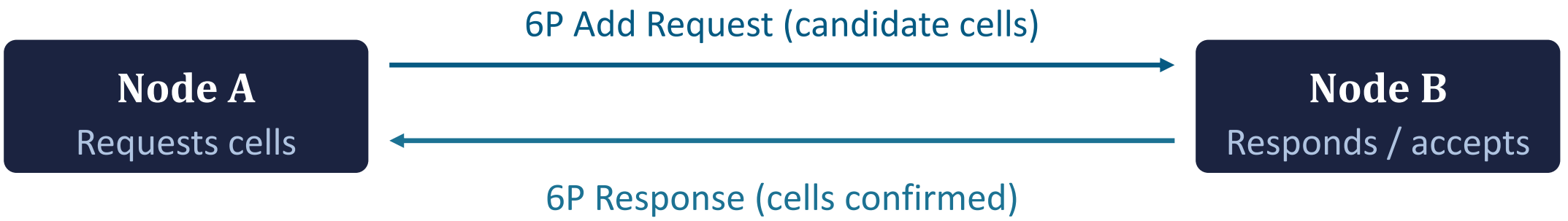
TSCH — Time-Slotted Channel Hopping



6TISCH

6top & the 6P Protocol

The 6top sublayer sits between IPv6 and TSCH. Neighbors use the 6P protocol to add, delete or relocate schedule cells — driven by a pluggable Scheduling Function (SF).



ADD

Reserve new cells for growing traffic demand

DELETE

Free cells no longer needed

RELOCATE

Move a cell if it suffers interference

6TISCH

Scheduling Approaches & Security

Who Builds the Schedule?

Static
Fixed schedule pre-configured; simplest, least adaptive.

Neighbor-to-Neighbor
Distributed: nodes negotiate cells via 6P (e.g. MSF).

Centralized (PCE)
A Path Computation Element schedules the whole mesh.

Keeping the Mesh Secure

- **Constrained Join Protocol (CoJP)**
  New nodes securely join via a Join Registrar/Coordinator.
- **Pre-shared network key**
  Bootstraps trust before per-link keys are issued.
- **Link-layer security**
  AES-128 protects frames exchanged over TSCH cells.
- **Rejoin & key rotation**
  Periodic key refresh limits exposure over time.

NEIGHBOR DISCOVERY

IPv6 Neighbor Discovery Protocol (NDP)

NDP runs over ICMPv6 and lets IPv6 nodes discover routers, resolve link-layer addresses, and configure themselves automatically — the IPv6 replacement for ARP.

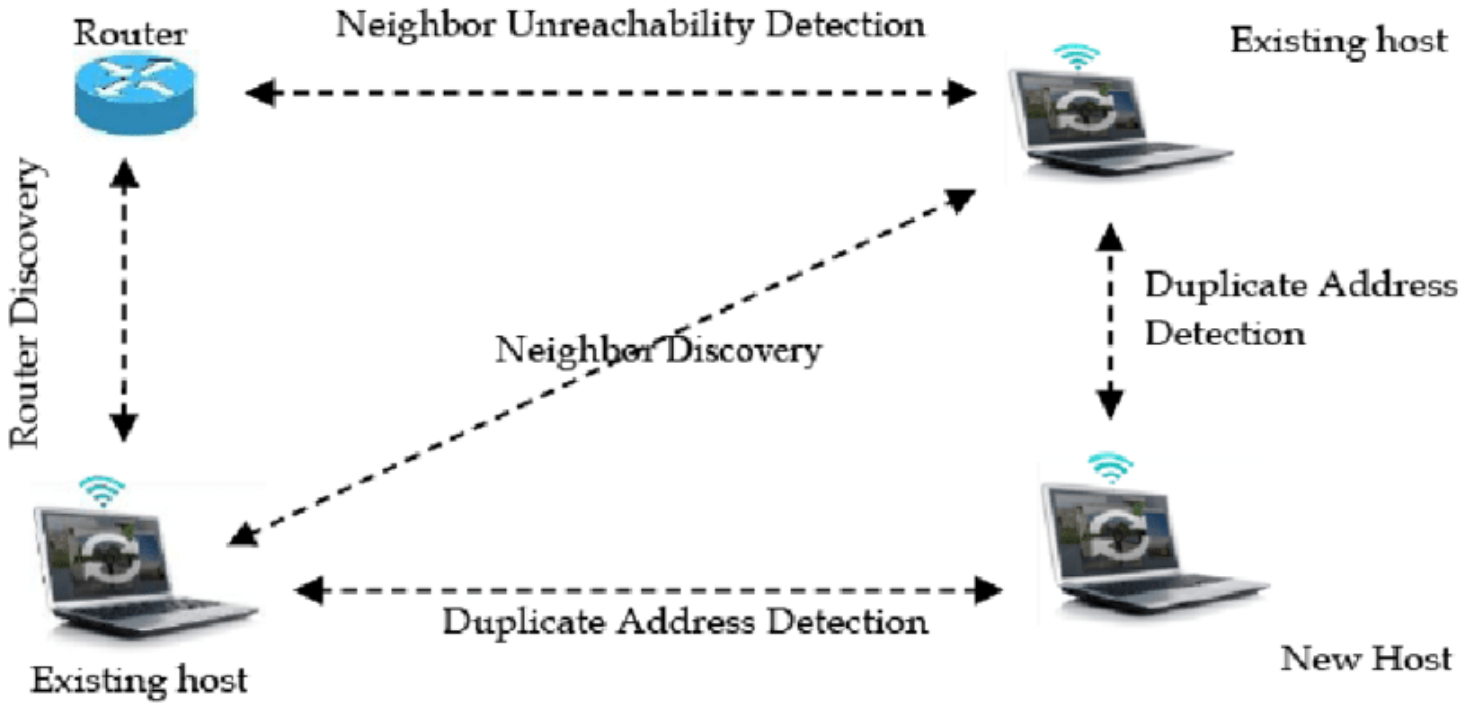
RS	RA	NS	NA	RD
Router Solicitation	**Router Advertisement**	**Neighbor Solicitation**	**Neighbor Advertisement**	**Redirect**
Host asks: "any routers here?"	Router announces prefixes & parameters	Resolve an address / verify reachability	Reply carrying the link-layer address	Router points host to a better next hop

Also handles: Address Autoconfiguration (SLAAC) · Duplicate Address Detection (DAD) · Neighbor Unreachability Detection (NUD)

Neighbour Discovery

NDP (Neighbor Discovery Protocol) Operations

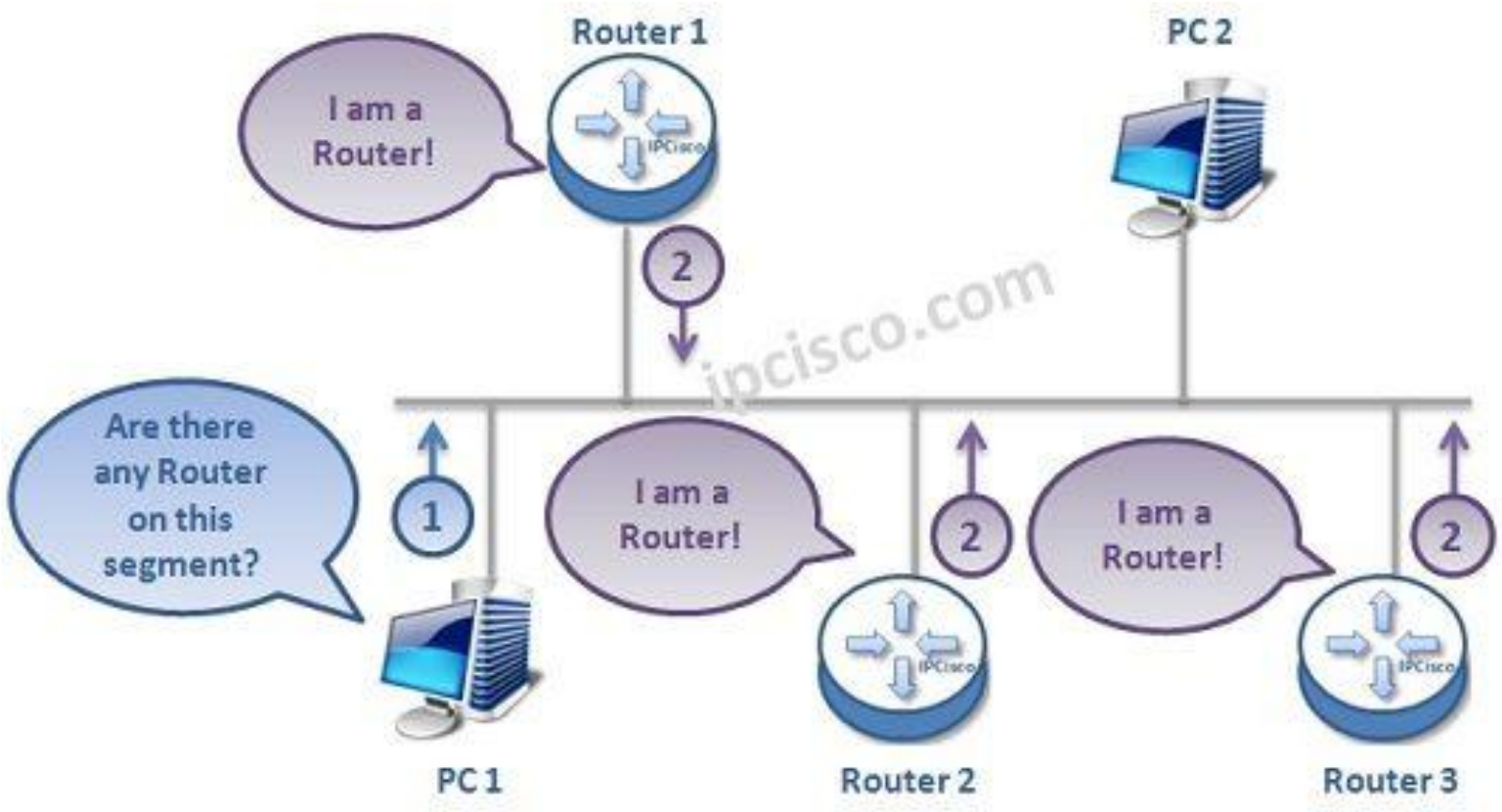
- Router Discovery
- Router Advertisements
- Neighbour Discovery
- Neighbour Advertisements
- Next-Hop Redirection
- Duplicate Address Detection



Neighbour Discovery

NDP (Neighbor Discovery Protocol) Operations

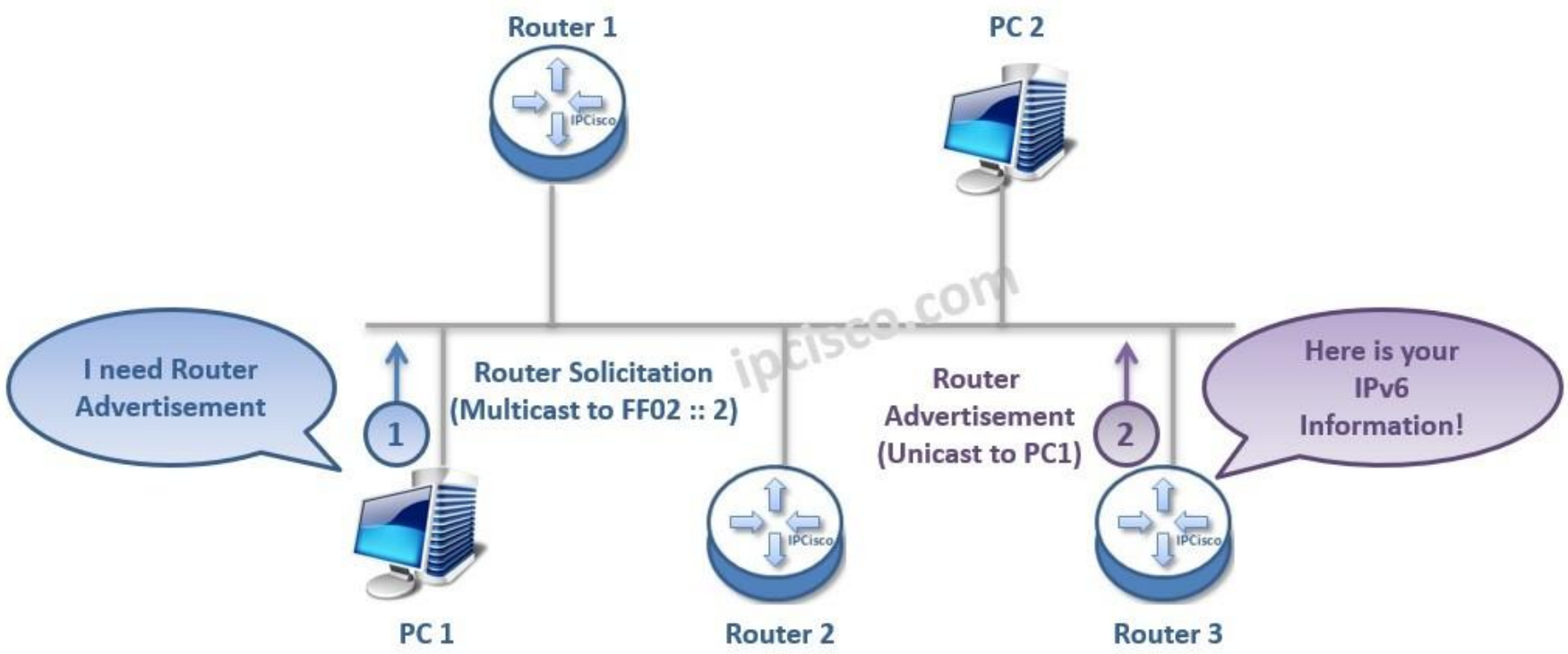
- Router Discovery



Neighbour Discovery

NDP (Neighbor Discovery Protocol) Operations

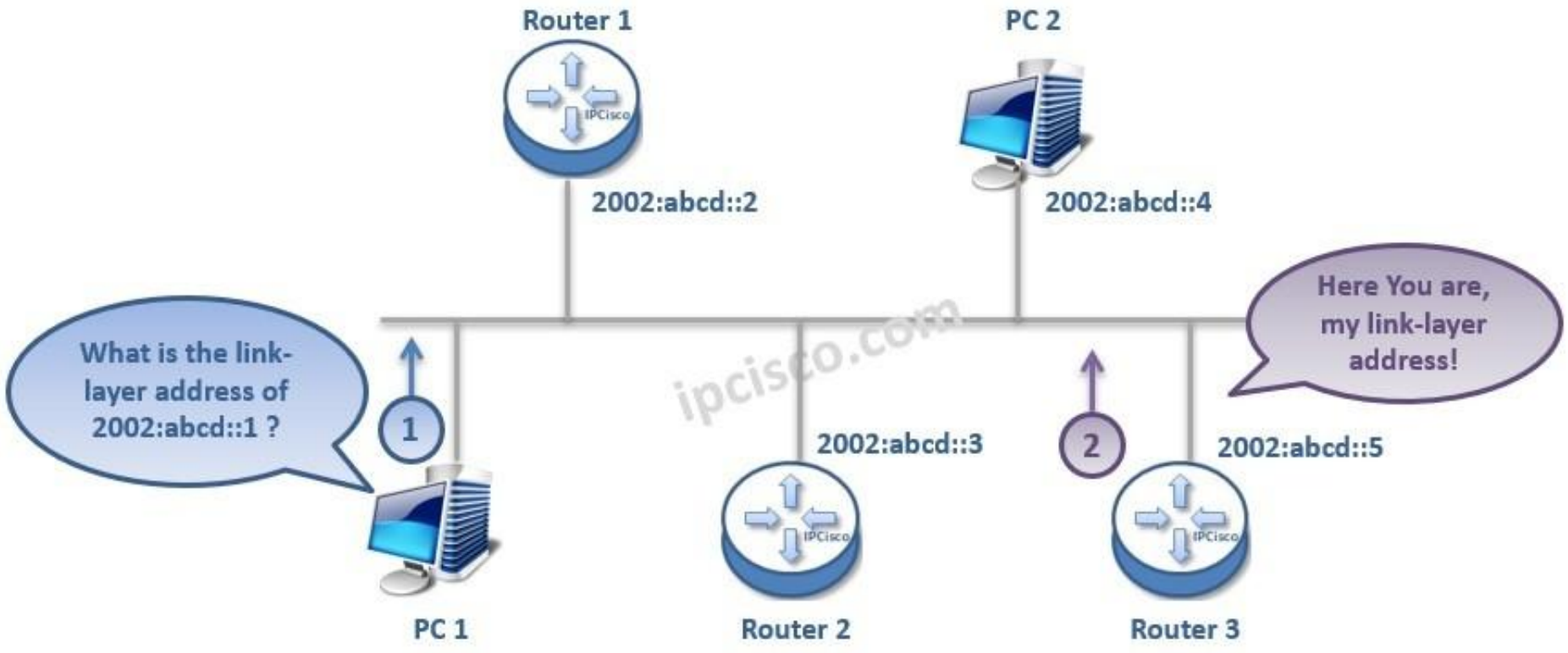
- Router Advertisements



Neighbour Discovery

NDP (Neighbor Discovery Protocol) Operations

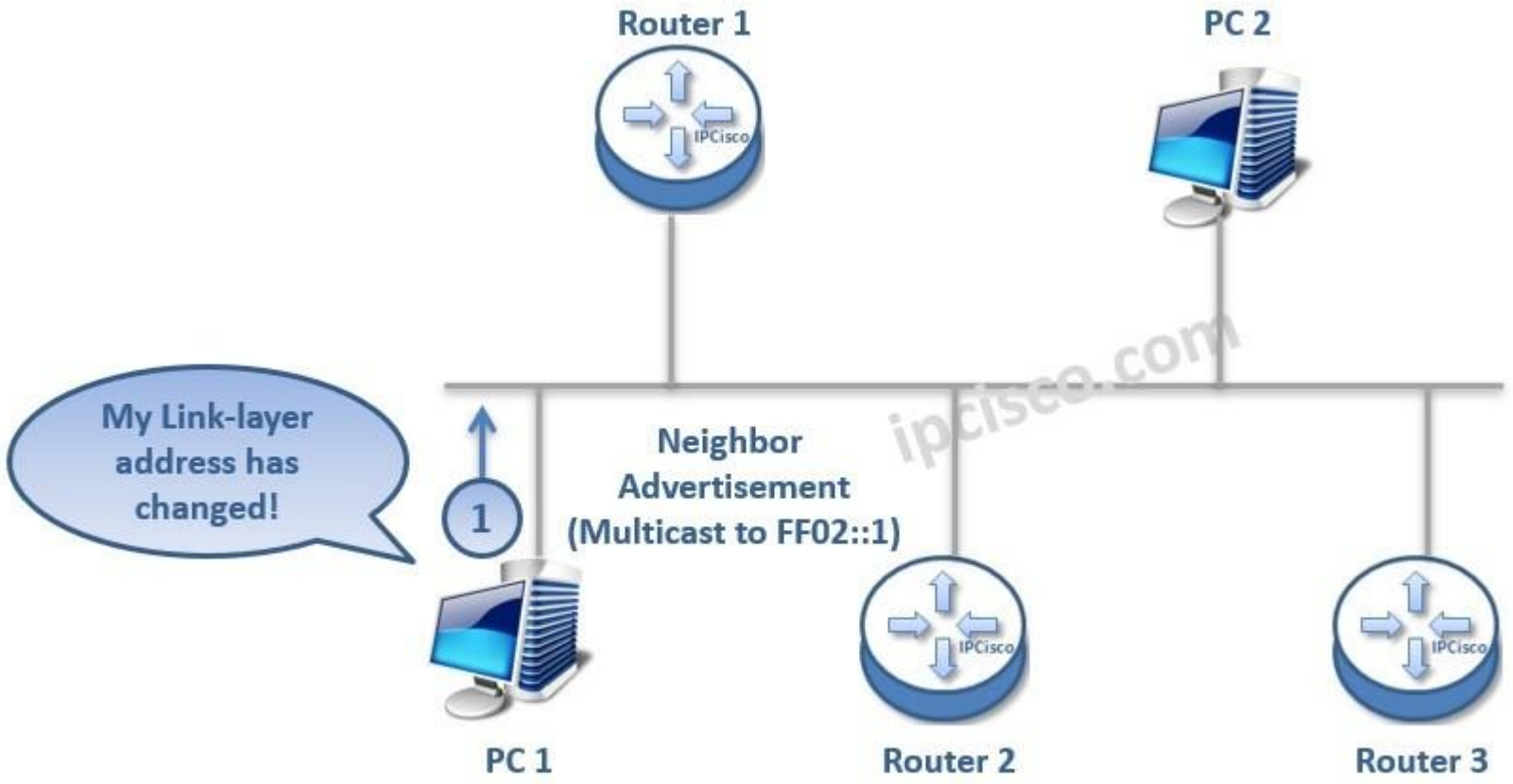
- Neighbour Discovery



Neighbour Discovery

NDP (Neighbor Discovery Protocol) Operations

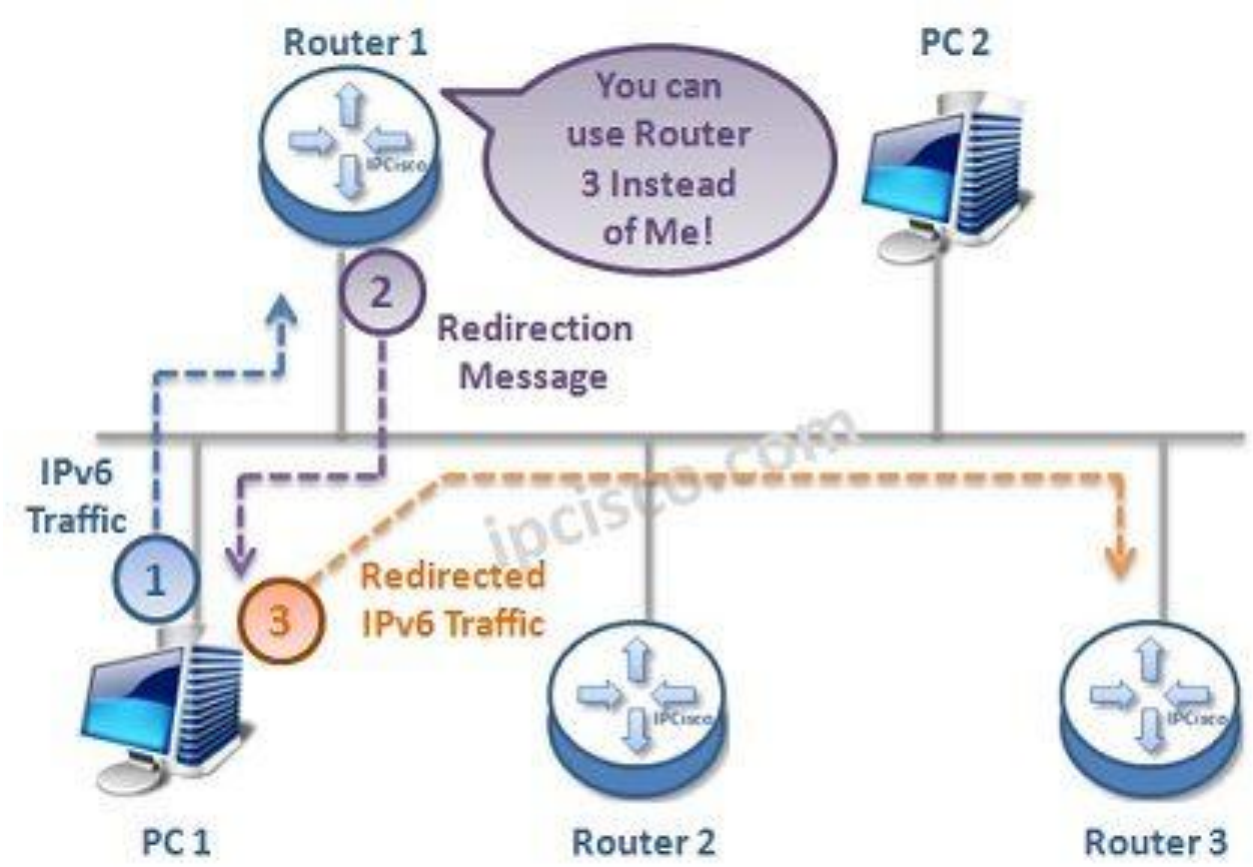
- Neighbour Advertisements



Neighbour Discovery

NDP (Neighbor Discovery Protocol) Operations

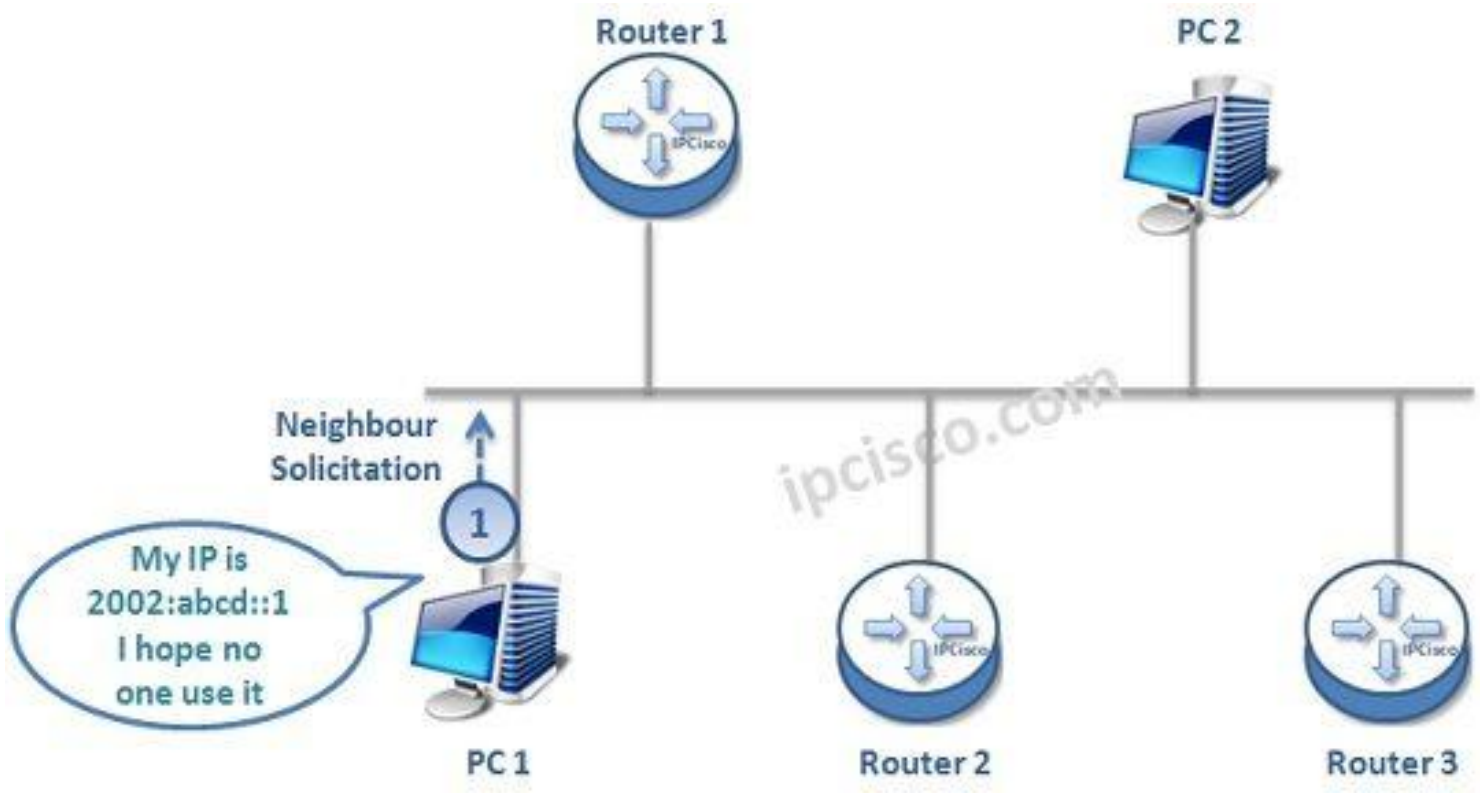
- Next-Hop Redirection



Neighbour Discovery

NDP (Neighbor Discovery Protocol) Operations

- Duplicate Address Detection



NEIGHBOR DISCOVERY

Why Classic ND Breaks Down in LLNs

Multicast Is Expensive

RS/RA and NS/NA rely on multicast — costly to send and to wake up for on battery-powered radios.

Assumes a Shared Link

Classic ND assumes every node hears every multicast; multihop mesh links are partial and asymmetric.

No Persistent State

Periodic Neighbor Unreachability Detection re-verifies neighbors repeatedly, burning energy.

Designed for Ethernet/Wi-Fi

NDP targets shared broadcast media, not multihop, low-bandwidth radio meshes.

BRINGING IT TOGETHER

6TiSCH and Neighbor Discovery, End to End

1

Secure Join (CoJP)

New node authenticates and receives keys plus initial TSCH cells.

2

Synchronize on TSCH

Node aligns to the slotframe; a shared cell carries control traffic.

3

6LoWPAN ND Registration

Node sends unicast NS+ARO to register its address with its 6LR.

4

Join the RPL DODAG

Node picks a preferred parent and gets an IPv6 route to the border router.

Result: a node that is time-synchronized, securely joined, individually addressable, and routable — all through the network layer.

SUMMARY

Key Takeaways

- IPv6 over TSCH for deterministic, low-power meshes
- 6top / 6P negotiates schedule cells between neighbors
- Static, distributed or centralized (PCE) scheduling
- CoJP secures the join process

- Classic NDP: RS, RA, NS, NA, Redirect over ICMPv6
- Multicast-heavy design assumes shared broadcast links
- Breaks down over lossy, multihop, constrained meshes

- Unicast, host-initiated registration (RFC 6775 / 8505)
- ARO carries identity + lifetime; 6CO shares context
- DAR/DAC lets a border router prevent duplicate addresses

REFERENCES

Further Reading & Questions

RFC 8180 / 8480	Minimal 6TiSCH Configuration · 6top Protocol (6P)	**RFC 9033**	6TiSCH Architecture
IEEE 802.15.4e	TSCH mode: time-slotted channel hopping MAC	**RFC 4861**	Neighbor Discovery for IP version 6 (IPv6)
RFC 6775 / 8505	6LoWPAN Neighbor Discovery & extensions (EARO)	**RFC 9010**	Routing for RPL Leaves (registration-based next hop)

DHCP , ICMP

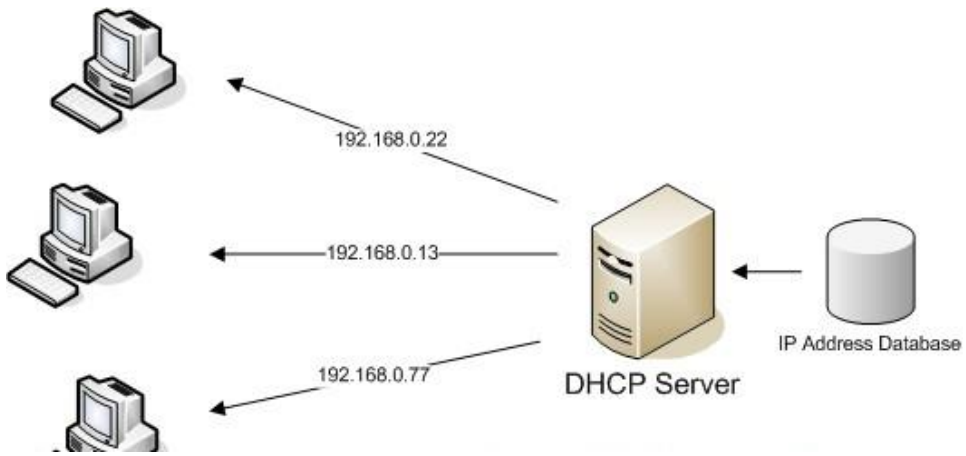
1. Network Layer in the IoT Stack
2. DHCP — Concepts & Operation
3. DHCP for Constrained IoT Networks
4. ICMP — Concepts & Message Types
5. ICMPv6 & Neighbor Discovery in IoT
6. Comparison & Key Takeaways

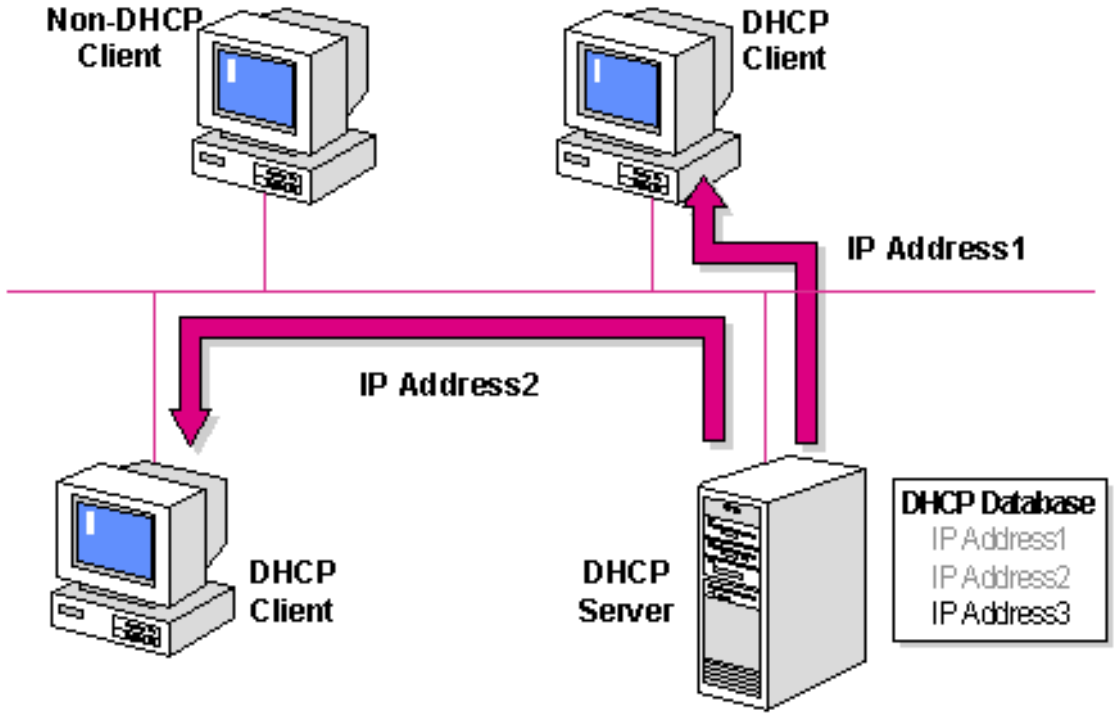
FOUNDATIONS

The Network Layer in the IoT Stack

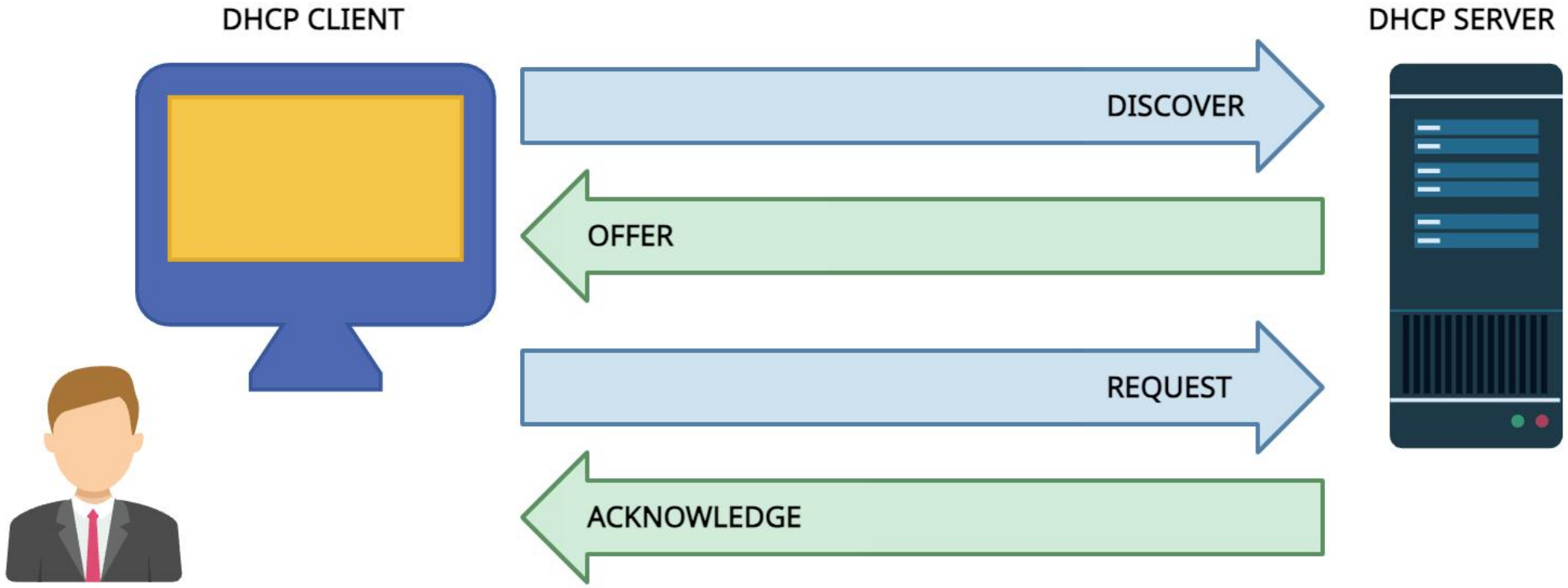
Application Layer	CoAP · MQTT · HTTP
Transport Layer	UDP · TCP
Network Layer	IPv6 · 6LoWPAN · RPL · DHCP · ICMP
Link Layer	IEEE 802.15.4 · BLE · LoRa

DHCP


192.168.0.22
192.168.0.13
192.168.0.77
DHCP Server
IP Address Database


Non-DHCP Client
DHCP Client
IP Address1
IP Address2
DHCP Client
DHCP Server
DHCP Database
IP Address1
IP Address2
IP Address3

How DHCP Works?



MOTIVATION

Why the Network Layer Matters in IoT

A **Addressing**

Every sensor and actuator needs a unique, manageable IP address to join the network.

D **Diagnostics**

Faults must be detected quickly across low-power, lossy wireless links.

S **Scalability**

Thousands of devices join, leave, and rejoin dynamically — configuration must be automatic.

DHCP BASICS

DHCP — What & Why

IP

Dynamic Host Configuration Protocol

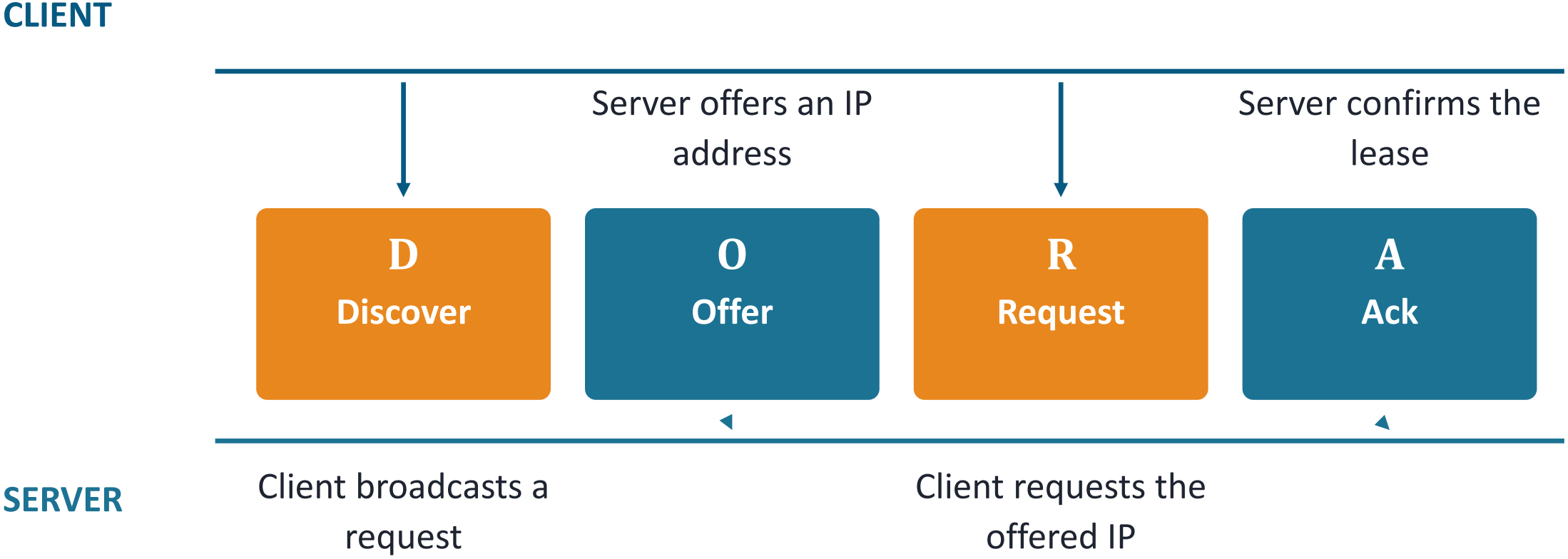
Automatically assigns IP addresses and network settings to a device the moment it joins the network — no manual setup required.

DHCP Assigns:

- IP Address
- Subnet Mask
- Default Gateway
- DNS Server
- Lease Duration

DHCP OPERATION

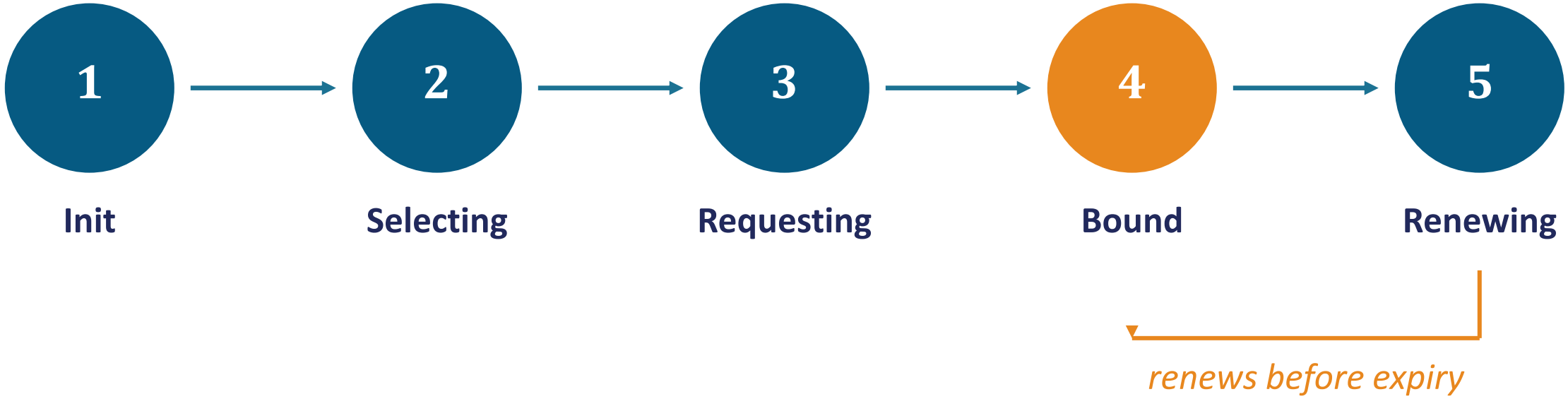
The DORA Process



D – O – R – A completes in a single, automatic handshake between client and server.

DHCP LEASE

The DHCP Lease Lifecycle



Renewal loop: the device periodically renews its lease before expiry, cycling between Bound and Renewing.

DHCP IN PRACTICE

DHCP for Constrained IoT Networks

Stateless preferred

Constrained devices often favor SLAAC over full DHCPv6 to save energy and memory.

DHCPv6 for control

Stateful DHCPv6 is used when centralized address and prefix control is required.

Prefix Delegation (PD)

DHCPv6-PD delegates address blocks to 6LoWPAN border routers automatically.

Lightweight relays

Compact DHCP relay agents forward requests from low-power mesh nodes to the server.

ICMP BASICS

ICMP — What & Why

!

Internet Control Message Protocol

Reports errors and provides network diagnostics between devices — it does not carry user application data.

Key Facts:

- Runs directly over IP — no port number
- Powers tools like ping and traceroute
- Critical for fault detection in IoT mesh networks

ICMP

ICMP (Internet Control Message Protocol)

- **ICMP** is a network protocol used primarily for diagnostic and error reporting in IP networks.
- Unlike protocols that carry user data (like TCP or UDP), ICMP is designed to send messages about the status of the network itself.

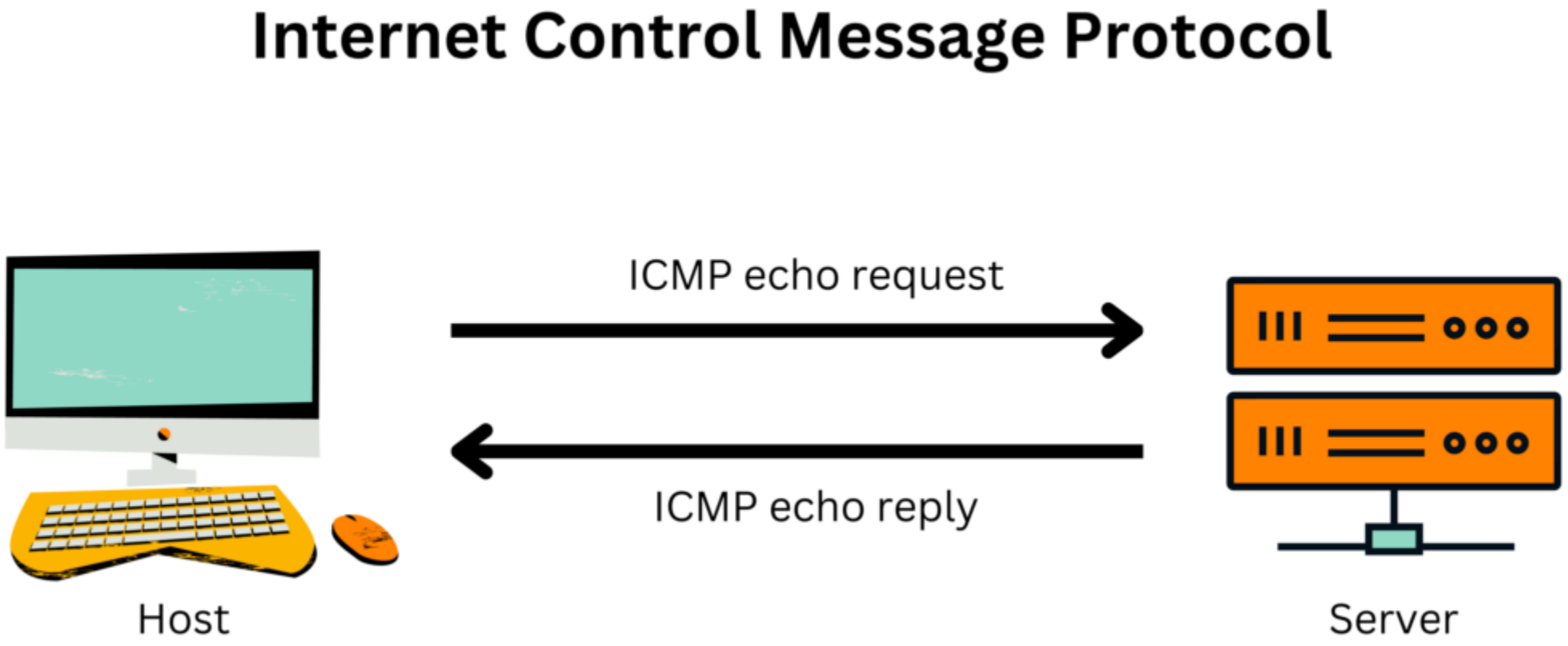


ICMP

ICMPv6 Enhancements

- Defined in **RFC 4443-by IETF**
- Integrates Neighbor Discovery Protocol (NDP) functions, replacing ARP.
- Common messages:
  - Router Solicitation (Type 133)
  - Router Advertisement (Type 134)
  - Neighbour Solicitation (Type 135)
  - Neighbour Advertisement (Type 136)
  - Redirect (Type 137)

ICMP


Internet Control Message Protocol
ICMP echo request
ICMP echo reply
Host
Server

ICMP MESSAGES

Key ICMP Message Types

0 / 8 — **Echo Reply / Request**

Reachability test — the basis of ping

3 — **Destination Unreachable**

Host, network, or port could not be reached

11 — **Time Exceeded**

TTL expired — used by traceroute

5 — **Redirect**

Informs a host of a better next-hop route

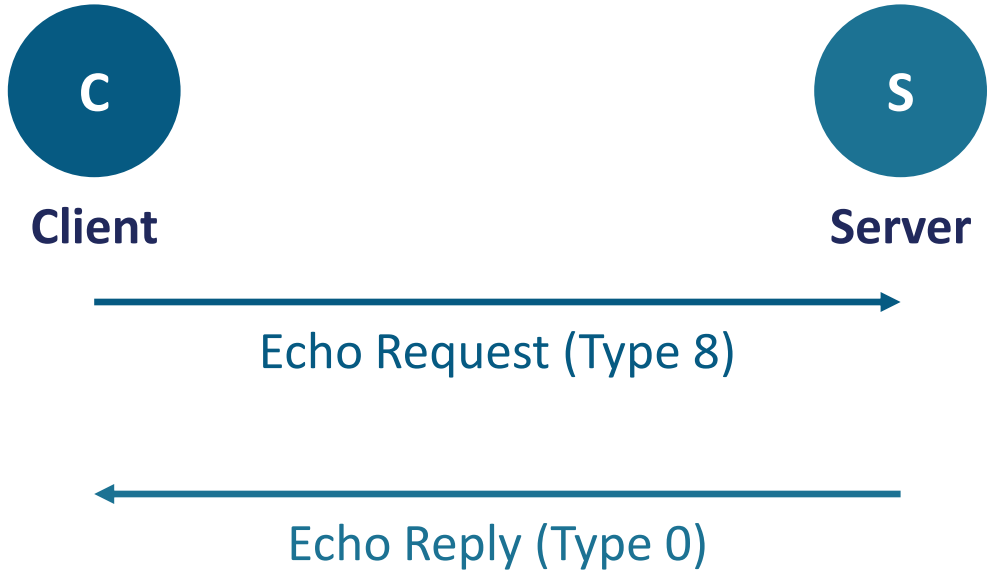
ICMP PACKET

ICMP Packet Fields & the Ping Exchange

Packet Fields

- Type
- Code
- Checksum
- Identifier
- Sequence Number

Ping Exchange



ICMPv6 & Neighbor Discovery in IoT

Router Solicitation / Advertisement

Devices discover the network prefix and default router on join.

Neighbor Solicitation / Advertisement

Resolves link-layer addresses, replacing IPv4 ARP.

6LoWPAN Autoconfiguration

NDP is the backbone of stateless address configuration for sensor nodes.

RPL Topology Support

Assists duplicate address detection and topology maintenance in mesh routing.

SUMMARY

Key Takeaways

1. DHCP automates address assignment; ICMP handles error reporting and diagnostics.
2. Both operate at or near the network layer and are essential for IoT scalability.
3. IoT adapts each protocol for constrained devices — DHCPv6-PD for gateways, ICMPv6 NDP for sensor autoconfiguration.
4. Reliable addressing and fault detection together keep large, dynamic IoT networks healthy.

ROADMAP

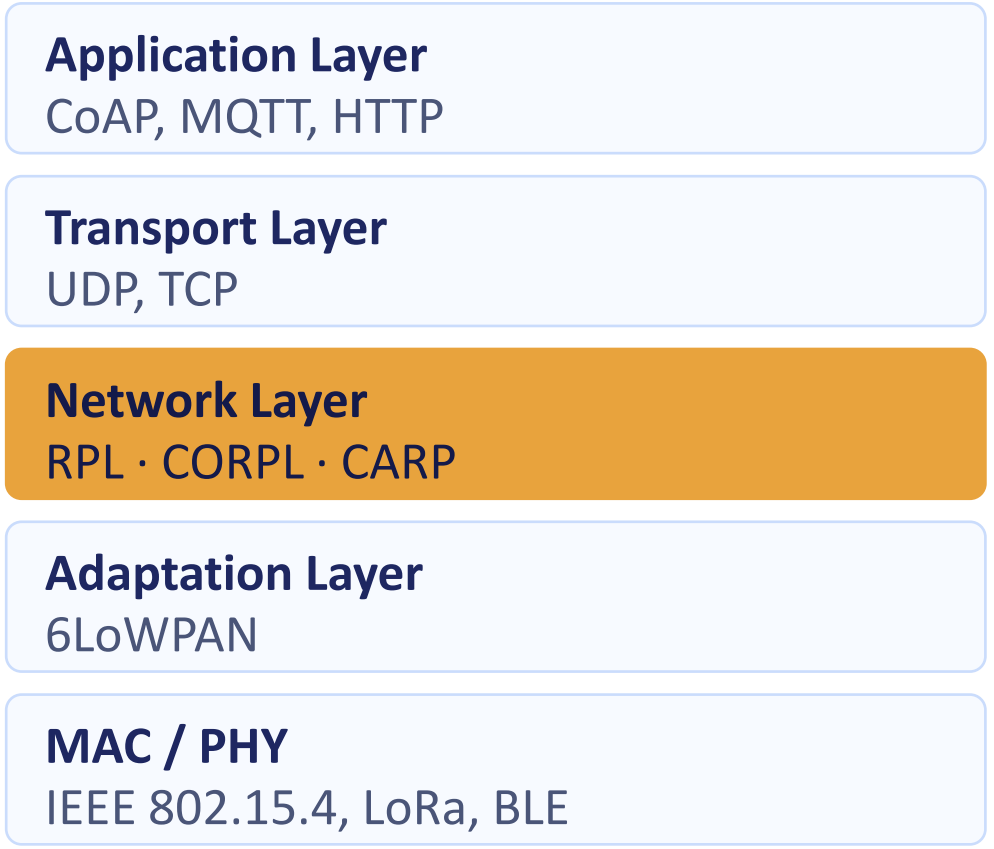
RPL, CORPL, CARP

1. Recap — the Network Layer in IoT
2. Why LLNs need purpose-built routing
3. RPL — Routing Protocol for Low-Power and Lossy Networks
4. CORPL — Cognitive RPL for cognitive radio networks
5. CARP — Channel-Aware Routing Protocol for underwater networks
6. Comparative summary and key takeaways

RECAP

The Network Layer in the IoT Stack

- Handles addressing and routing of packets across constrained IoT nodes and toward the gateway.
- Sits above 6LoWPAN and below the transport layer in the IoT stack.
- — RPL, CORPL, CARP — are routing solutions built specifically for this layer.



MOTIVATION

Why Low-Power, Lossy Networks Need New Routing

1

Constrained Devices

Limited battery, memory and CPU on sensor nodes.

2

Lossy Links

High packet loss and unstable wireless links.

3

Dynamic Topology

Nodes join, leave, sleep or move frequently.

4

Skewed Traffic

Mostly many-to-one (sensors → sink) flows.

Classical protocols such as OSPF and RIP assume stable, resource-rich links — they do not fit these constraints. This motivates RPL, CORPL and CARP.

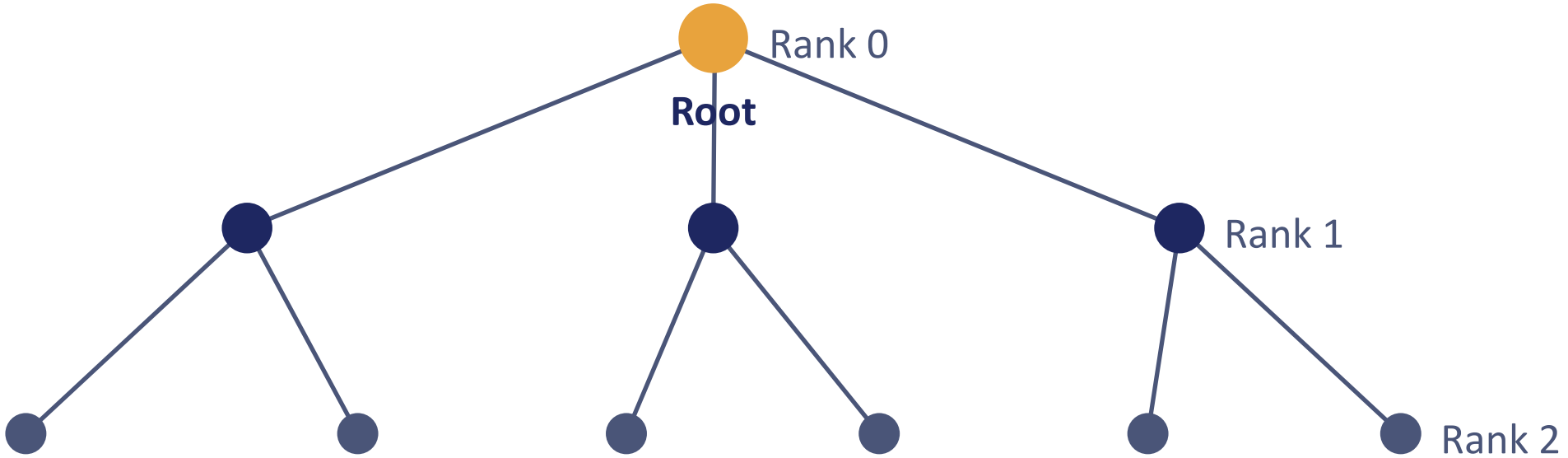
PROTOCOL 1 OF 3

RPL — Routing Protocol for LLNs

- Defined by the IETF ROLL working group in RFC 6550, for IPv6-based Low-power and Lossy Networks.
- A proactive, distance-vector style protocol built around a graph rather than a simple tree.
- Organizes the network into a Destination-Oriented Directed Acyclic Graph (DODAG).
- Supports three traffic patterns: multipoint-to-point, point-to-multipoint, and point-to-point.

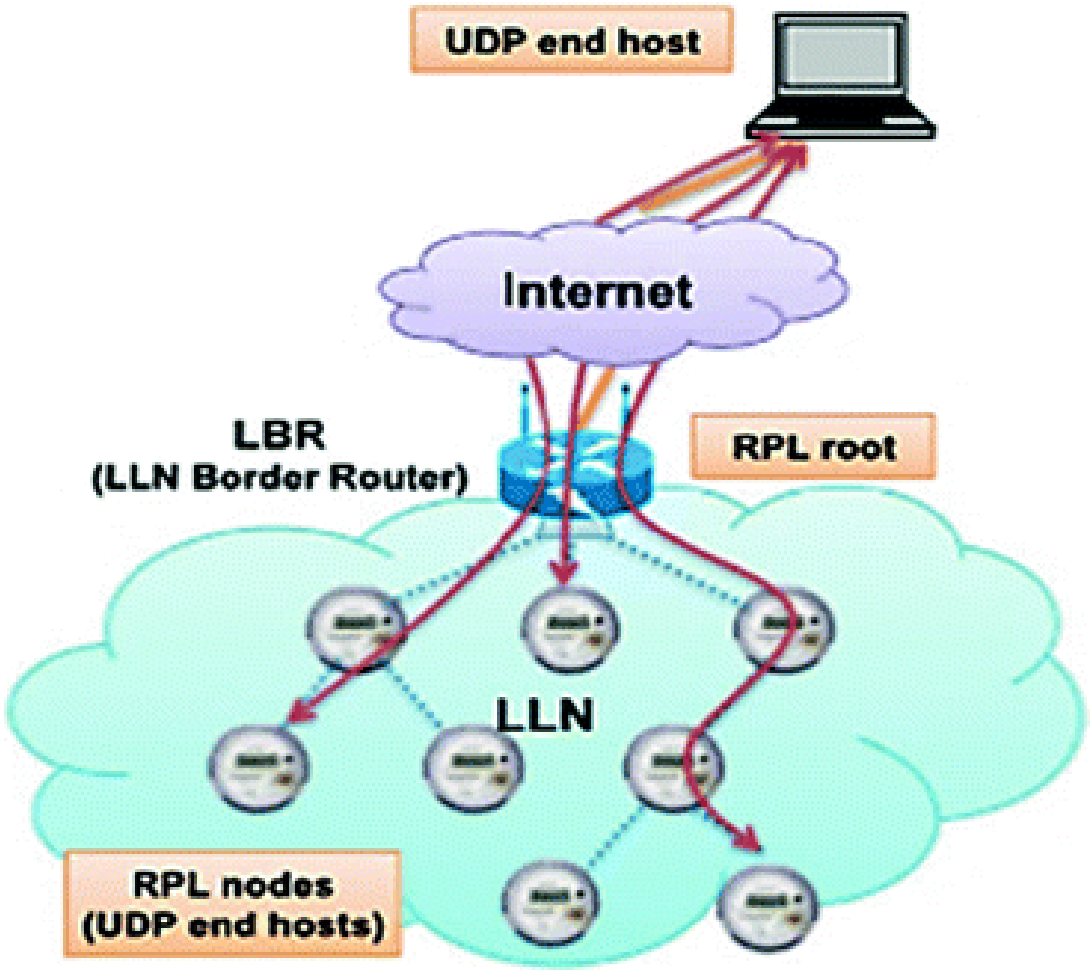
RPL STRUCTURE

The DODAG: Destination-Oriented DAG



- Root (sink / border router) is the DODAG destination; rank grows with distance from it.
- Each node picks a preferred parent toward the root, forming loop-free upward routes.
- The graph self-heals: nodes re-select parents when links fail.

RPL


UDP end host
Internet
LBR
(LLN Border Router)
RPL root
LLN
RPL nodes
(UDP end hosts)

RPL MECHANICS

Control Messages & Objective Function

DIO

DODAG Information Object

Sent downward from parents; builds and maintains the DODAG and advertises rank.

DAO

Destination Advertisement Object

Sent upward by children; builds downward routes toward leaf nodes.

DIS

DODAG Information Solicitation

A node requests a DIO when it needs to join or refresh the graph.

Objective Function (OF): computes a node's rank and selects its preferred parent using a routing metric — e.g. OF0 (hop count) or MRHOF (Expected Transmission Count, ETX).

RPL ASSESSMENT

Advantages & Limitations

Advantages

- Standardized by the IETF (RFC 6550)
- Scales to large, sparse LLN deployments
- Supports multiple traffic patterns
- Self-healing via re-parenting on link failure

Limitations

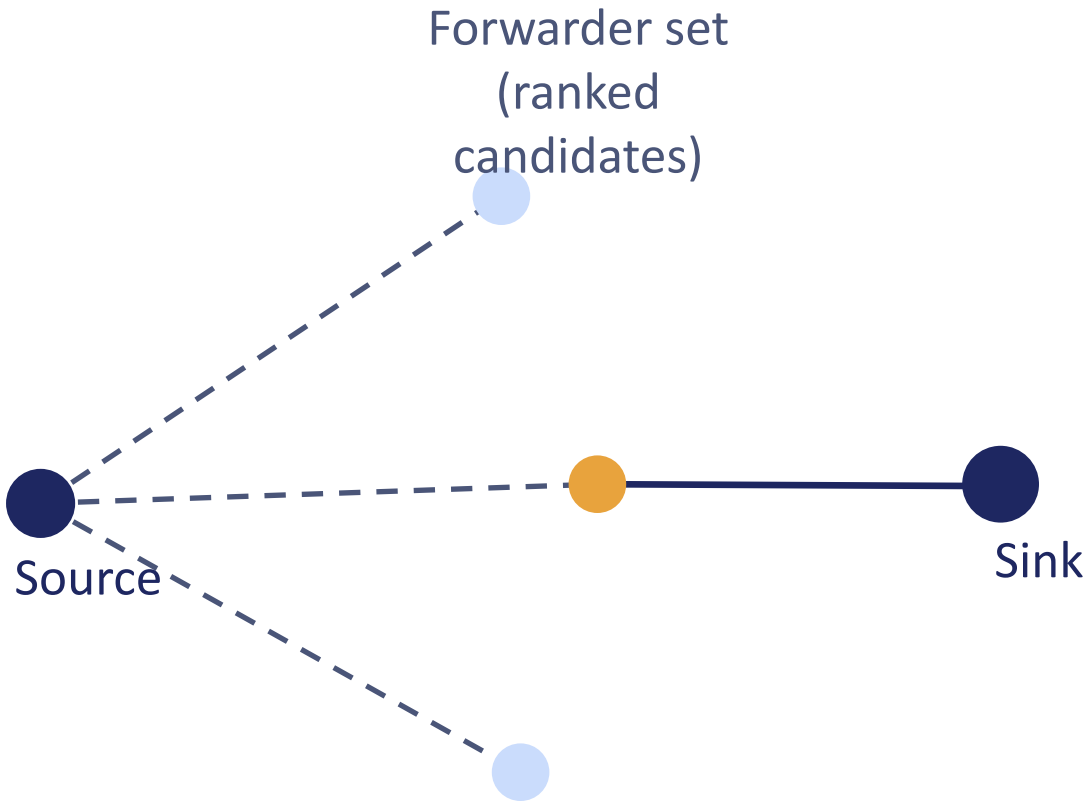
- Control-message overhead (DIO/DAO traffic)
- Slow convergence after topology change
- Not designed for mobile or spectrum-varying nodes
- Storing mode needs extra memory for routing tables

PROTOCOL 2 OF 3

CORPL — Cognitive RPL

- An extension of RPL proposed for Cognitive Radio Ad-hoc Networks (CRAHNs).
- Cognitive nodes opportunistically use free spectrum bands — available channels change over time.
- CORPL keeps RPL's DODAG concept but adapts it to handle this constantly changing topology.
- Goal: maintain reliable routes even as spectrum availability shifts node connectivity.

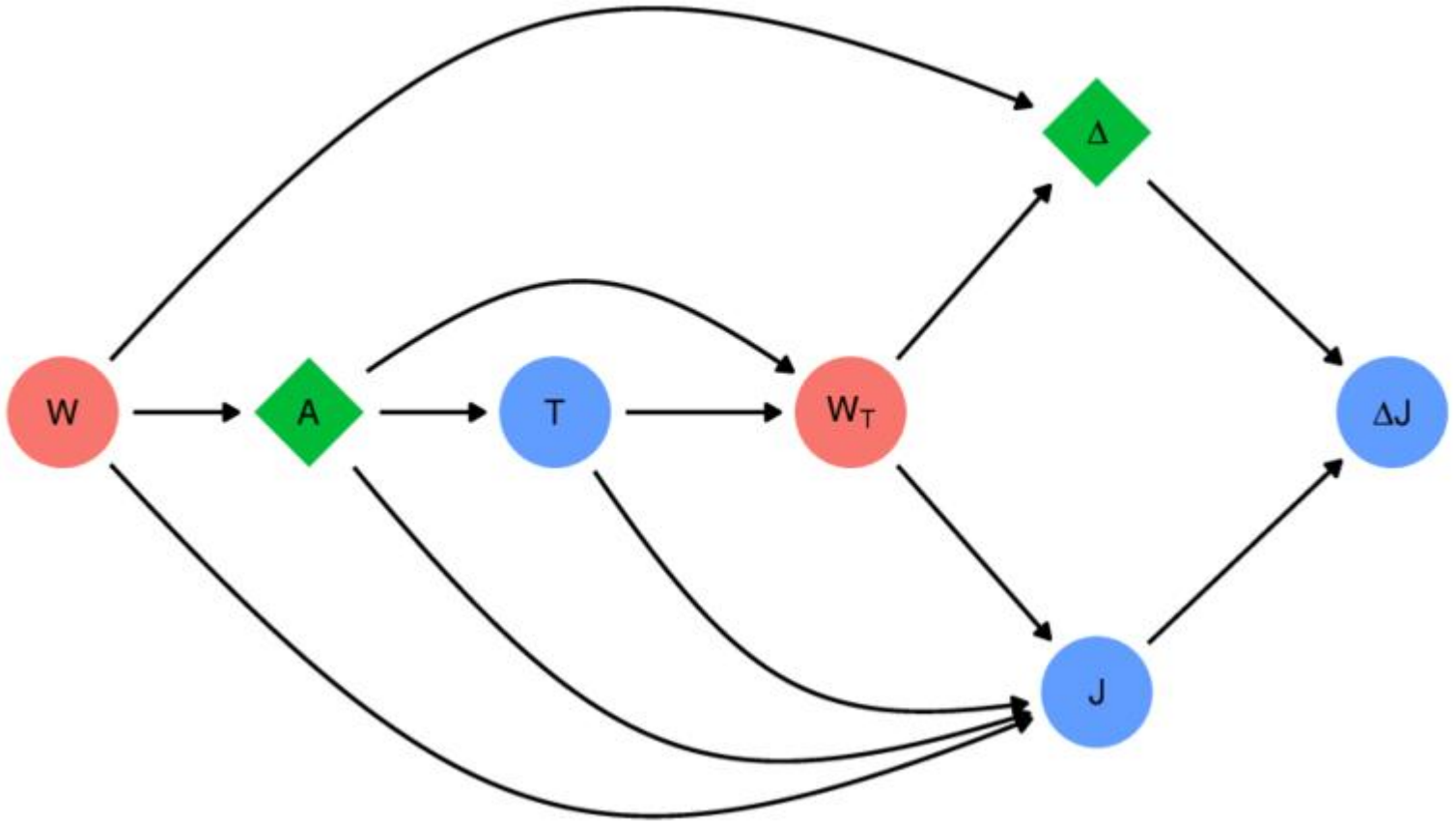
Opportunistic Forwarding in CORPL



- Each node keeps a neighbor table with delivery probabilities to candidate next hops.
- Instead of one fixed parent, packets go to a forwarder set; the best-placed node relays.
- Rank reflects both RPL-style distance AND current spectrum / link availability.

CORPL

Directed Acyclic Graph-Example
No Loop exist

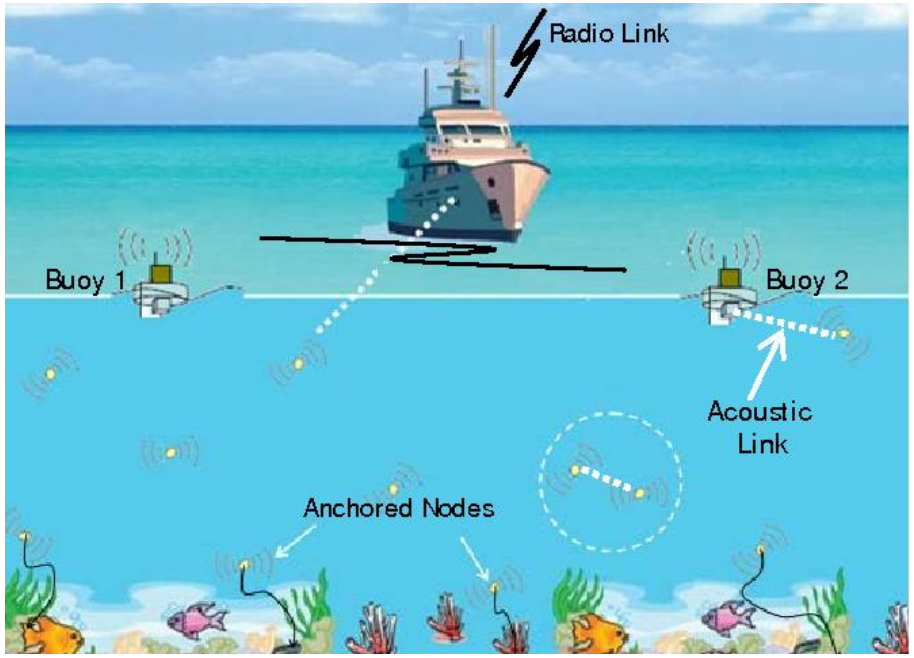


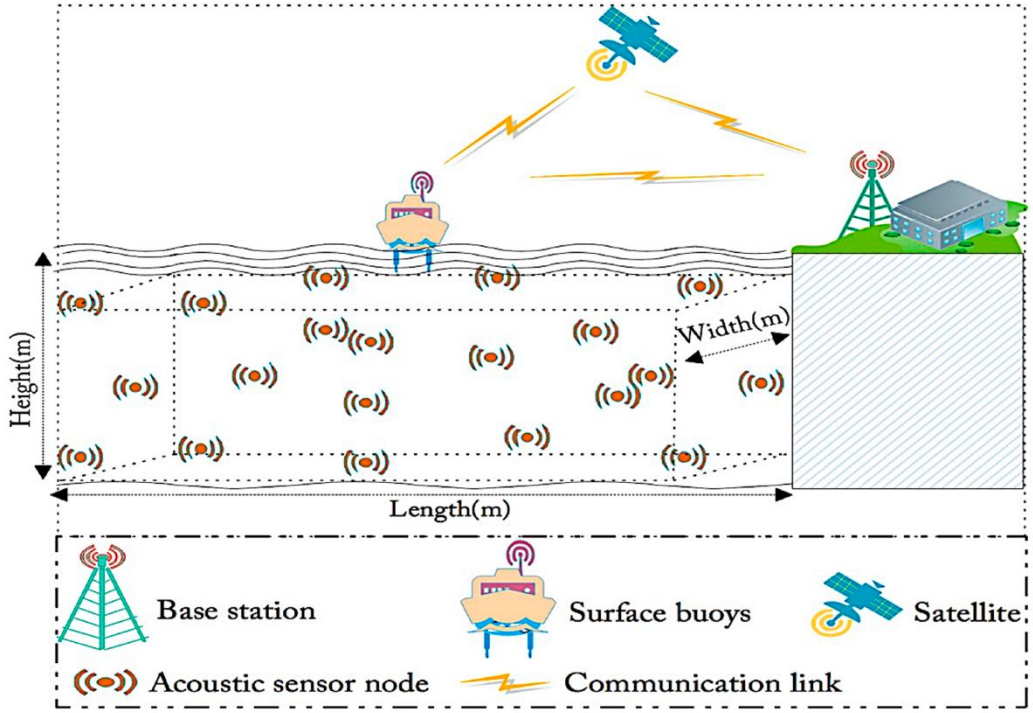
PROTOCOL 3 OF 3

CARP — Channel-Aware Routing Protocol

- Designed for Underwater Wireless Sensor Networks (UWSNs), which use acoustic communication.
- Underwater channels suffer long propagation delay, multipath fading and high, variable loss.
- CARP is distributed and hop-by-hop: it picks the next forwarder based on historical link quality rather than a pre-built graph.
- No global topology is maintained — decisions are made locally, hop by hop, toward the sink.

CARP


Radio Link
Buoy 1
Buoy 2
Acoustic
Link
Anchored Nodes


Height(m)
Width(m)
Length(m)
Base station
Surface buoys
Satellite
Acoustic sensor node
Communication link

CARP MECHANICS

The HELLO – PING – PONG Handshake

1

HELLO (setup)

Sink broadcasts a HELLO; it propagates hop-by-hop so every node learns its hop count to the sink.

2

PING

A source node with data broadcasts a PING to its neighbors closer to the sink.

3

PONG

Each neighbor replies with a PONG carrying its link-quality history (past successful transmissions).

4

Forwarder Selection

The source picks the neighbor with the best channel quality as the next-hop forwarder.

SYNTHESIS

RPL vs. CORPL vs. CARP

RPL Terrestrial LLNs	CORPL Cognitive Radio Networks	CARP Underwater Networks
IPv6 sensor / IoT networks	Cognitive radio ad-hoc networks	Underwater acoustic sensor networks
DODAG (tree-like graph)	DODAG + ranked forwarder sets	No fixed graph — hop-by-hop
Rank via ETX / hop count (OF)	Delivery probability + spectrum state	Historical link-quality metric
DIO / DAO / DIS messaging	Opportunistic multi-forwarder relay	HELLO / PING / PONG handshake
Best for static-ish topologies	Handles dynamic channel changes	Handles long delay & high loss

KEY TAKEAWAYS

Same Problem, Three Environments

- **RPL** — The IETF-standard foundation for IPv6 routing in terrestrial low-power, lossy IoT networks.
- **CORPL** — Extends RPL's DODAG with opportunistic, forwarder-set routing to survive shifting spectrum in cognitive radio networks.
- **CARP** — A purpose-built, hop-by-hop protocol that copes with the long delay and heavy loss of underwater acoustic links.

Common thread: adapt routing decisions to constrained, lossy, and dynamic link conditions.

WRAP-UP

References & Questions

Key References

- RFC 6550 — RPL: IPv6 Routing Protocol for LLNs, IETF ROLL WG
- T. Winter et al., “RPL: IPv6 Routing Protocol for Low-Power and Lossy Networks,” IETF, 2012
- CORPL and CARP — as discussed in IoT network-layer routing literature

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Questions & Discussion

Thank you for your attention.